

Write a python program which accept a word a line of text and display its characters by using while loop

```
In [2]: word=input("Enter a word or line of txt:") # word Python
        index=0 # Initliation part
        while(index<len(word)):
            print(word[index])
            index=index+1
```

P
y
t
h
o
n

```
In [4]: word=input("Enter a word or line of txt:") # word Python
        index=0 # Initliation part
        while(index<len(word)):
            print(word[index])
            index=index+1
```

M
a
h
a
b
o
o
b

K
h
a
n

```
In [6]: word=input("Enter a word or line of txt:") # word Python
        index=0 # Initliation part
        while(index<len(word)):
            print(word[index])
            index=index+1
```

M
R
I
I
R
S

```
In [ ]: ##### w.r.t -ve index with forward direction
```

```
In [10]: word=input("Enter a word or line of txt:") # word Python
          index=0 # Initliation part
          while(index<len(word)):
              print(word[index])
```

```

    index=index+1
else:
    print("-----OR-----")
    index=-len(word)
    while(index<=-1):
        print(word[index])
        index=index+1

```

M
A
H
A
B
O
O
B

K
H
A
N

-----OR-----

M
A
H
A
B
O
O
B

K
H
A
N

```

In [12]: word=input("Enter a word or line of txt:") # word Python
         index=0 # Initliation part
         while(index<len(word)):
             print(word[index])
             index=index+1
         else:
             # w.r.t +ve index Forward Direction

             print("-----OR-----")
             index=-len(word)
             while(index<=-1):
                 print(word[index])
                 index=index+1
             else: # w.r.t +ve index BACKWORD Direction
                 print("-----OR-----")
                 index=len(word)-1
                 while(index>=0):
                     print(word[index])
                     index=index-1

```

```

M
R
I
I
R
S

C
D
O
E
-----OR-----
M
R
I
I
R
S

C
D
O
E
-----OR-----
E
O
D
C

S
R
I
I
R
M

```

```

In [10]: # while loop (17:25)

word = input("Enter a word OR Line of Text: ") # word = Python

# write +ve index forward direction
index = 0 # Initialization part
while (index < len(word)):
    print(word[index])
    index = index + 1
else:
    # write -ve index forward direction
    print("-----OR-----")
    index = -len(word)
    while (index < 0):
        print(word[index])
        index = index + 1
    else:
        print("-----")

# w.r.t +ve index backward direction

```

```

index = len(word) - 1
while (index >= 0):
    print(word[index])
    index = index - 1
else:
    print("-----")

# w.r.t -ve index backward direction
index = -1
while (index >= -len(word)):
    print(word[index])
    index = index - 1

```

```

K
h
a
n
-----OR-----
K
h
a
n
-----
n
a
h
K
-----
n
a
h
K

```

```

In [3]: # 25:20 For Loop
word=input("Enter a word or Line of text:") #word = Python
print("w.r.t +ve index ---- Forward direction")
for val in word:
    print(val)
else:
    print("-"*50)

```

```

w.r.t +ve index ---- Forward direction
M
a
h
a
b
o
o
b

K
h
a
n
-----

```

```

In [21]: word = input("Enter a Word OR Line of Text: ")    # word = MRIIRS

print("w.r.t +Ve Index --- FORWARD DIRECTION")
for val in word:
    print(val)

else:
    print("-" * 50)
    print("w.r.t +Ve Index --- FORWARD DIRECTION")

    for index in range(0, len(word)):
        print(word[index])

    else:
        print("-" * 50)
        print("w.r.t +Ve Index --- BACKWARD DIRECTION")

        for index in range(len(word) - 1, -1, -1):
            print(word[index])

    else:
        print("-" * 50)
        print("w.r.t -Ve Index --- FORWARD DIRECTION")

        for index in range(-len(word), 0):
            print(word[index])

    else:
        print("-" * 50)
        print("w.r.t -Ve Index --- BACKWARD DIRECTION")

        for index in range(-1, -len(word) - 1, -1):
            print(word[index]) #34:54
        else:
            print("-" * 50)
            print("w.r.t for loop --- BACKWARD DIRECTION")
            for val in word[::-1]:
                print(val)

```

```
w.r.t +Ve Index --- FORWARD DIRECTION
M
R
I
I
R
S
```

```
-----
w.r.t +Ve Index --- FORWARD DIRECTION
M
R
I
I
R
S
```

```
-----
w.r.t +Ve Index --- BACKWARD DIRECTION
S
R
I
I
R
M
```

```
-----
w.r.t -Ve Index --- FORWARD DIRECTION
M
R
I
I
R
S
```

```
-----
w.r.t -Ve Index --- BACKWARD DIRECTION
S
R
I
I
R
M
```

```
-----
w.r.t for loop --- BACKWARD DIRECTION
S
R
I
I
R
M
```

```
In [3]: #42:27 :- Python program to generate a multiplication table for a given number using
n=int(input("Enter the number :"))
if(n<=0):
    print("{} is invalid input:".format(n))
else:
    for i in range(1,11):
        print(i)
```

-1 is invalid input:

```
In [5]: #42:27 :- Python program to generate a multiplication table for a given number using
n=int(input("Enter the number :"))
if(n<=0):
    print("{} is invalid input:".format(n))
else:
    for i in range(1,11):
        print(i)
```

0 is invalid input:

```
In [7]: #42:27 :- Python program to generate a multiplication table for a given number using
n=int(input("Enter the number :"))
if(n<=0):
    print("{} is invalid input:".format(n))
else:
    for i in range(1,11):
        print(i)
```

1
2
3
4
5
6
7
8
9
10

```
In [11]: #42:27 :- Python program to generate a multiplication table for a given number using
n=int(input("Enter the number :"))
if(n<=0):
    print("{} is invalid input:".format(n))
else:
    print("-"*50)
    print("Multiplication table {}".format(n))
    print("-"*50)
    for i in range(1,11):
        print("\t{} x {} = {}".format(n,i, n*i))
    else:
        print("-"*50)
```

 Multible table 12

12 x 1 = 12
 12 x 2 = 24
 12 x 3 = 36
 12 x 4 = 48
 12 x 5 = 60
 12 x 6 = 72
 12 x 7 = 84
 12 x 8 = 96
 12 x 9 = 108
 12 x 10 = 120

In [15]: #42:27 :- Python program to generate a multiplication table for a given number using
 n=int(input("Enter the number :"))
 if(n<=0):
 print("{} is invalid input:".format(n))
 else:
 print("-"*50)
 print("Multible table {}".format(n))
 print("-"*50)
 for i in range(1,11):
 print("\t{} x {} = {}".format(n,i, n*i))
 else:
 print("-"*50)

-5 is invalid input:

In [17]: #42:27 :- Python program to generate a multiplication table for a given number using
 n=int(input("Enter the number :"))
 if(n<=0):
 print("{} is invalid input:".format(n))
 else:
 print("-"*50)
 print("Multible table {}".format(n))
 print("-"*50)
 for i in range(1,11):
 print("\t{} x {} = {}".format(n,i, n*i))
 else:
 print("-"*50)

0 is invalid input:

In [19]: #44:27 :- Python program to generate a multiplication table for a given number using
 n=int(input("Enter the number :"))
 if(n<=0):
 print("{} is invalid input:".format(n))
 else:
 print("-"*50)
 print("Multible table {}".format(n))
 print("-"*50)
 for i in range(1,11):
 print("\t{} x {} = {}".format(n,i, n*i))
 else:
 print("-"*50)

 Multible table 19

19 x 1 = 19
 19 x 2 = 38
 19 x 3 = 57
 19 x 4 = 76
 19 x 5 = 95
 19 x 6 = 114
 19 x 7 = 133
 19 x 8 = 152
 19 x 9 = 171
 19 x 10 = 190

```
In [25]: # 48:00 -> Programming find Sum of N natural numbers where N is +VE
n=int(input("Enter how many natural numbers sum do you want:"))
if(n<=0):
    print("{} is invalid inupt:".format(n))
else:
    print("-"*50)
    print("sum of natural numbers".format(n))
    s=0 # Initial sum called additive identity
    for i in range(1,n+1):
        print("\t{}".format(i))
        s+=i # OR s=s+i
    else:
        print("-"*50)
        print("sum={}".format(s))
        print("-"*50)
```

 sum of natural numbers

1
 2
 3
 4
 5

sum=15

```
In [27]: # 1:03:10 -> Programming find Sum of N natural numbers where N is +VE
n=int(input("Enter how many natural numbers sum do you want:"))
if(n<=0):
    print("{} is invalid inupt:".format(n))
else:
    print("-"*50)
    print("sum of natural numbers".format(n))
    s=0 # Initial sum called additive identity
    for i in range(1,n+1):
        print("\t{}".format(i))
        s+=i # OR s=s+i
    else:
        print("-"*50)
```

```
print("sum={}".format(s))
print("-"*50)
```

sum of natural numbers

```
1
2
3
4
5
6
7
8
9
10
```

sum=55

In [29]: *# 01:05:38 while Loop : Programming find Sum of N natural numbers where N is +VE?*
Program for Finding Sum of N Natural Numbers where N is +VE

```
n = int(input("Enter how many natural numbers sum you want: "))

if n <= 0:
    print("{} is Invalid Input".format(n))
else:
    print("-" * 50)
    print("Sum of {} Natural Numbers".format(n))
    print("-" * 50)

    s = 0    # Initial sum (Additive Identity)

    for i in range(1, n + 1):
        print("\t{}".format(i))
        s += i    # same as s = s + i

    print("-" * 50)
    print("Sum = {}".format(s))
    print("-" * 50)
```

```
-----  
Sum of 10 Natural Numbers  
-----  
    1  
    2  
    3  
    4  
    5  
    6  
    7  
    8  
    9  
   10  
-----  
Sum = 55  
-----
```

```
In [ ]: SESSION END
```